Q1. Solution

We are given a dataset of (x, y) points where $x \in R^+$. The goal is to choose the most appropriate feature map $\phi(x)$ that allows linear regression to model the observed nonlinear curve.

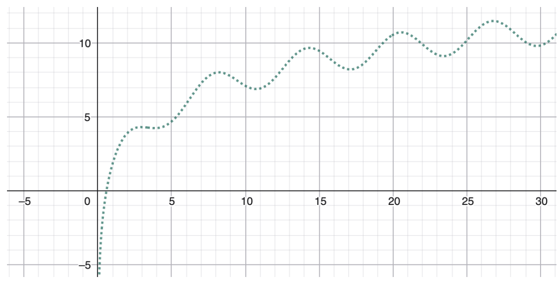

Figure 1: Observed data points

Understand Feature Mapping

Linear regression can model nonlinear relationships by transforming input features using a feature map:

$$\phi(x) : R \to R^N$$

This allows the model to remain linear in parameters while capturing nonlinear patterns in data.

Evaluate Each Feature Map Option

Option 1:

$$\phi(x) = [1, x, \sin(x)]$$

This captures linear growth and oscillations, but does not model logarithmic growth, which better matches the observed trend.

Option 2:

$$\phi(x) = [1, \sin(x)]$$

This only captures oscillatory behavior and ignores the increasing trend.

Option 3:

$$\phi(x) = [1, \log(x), \sin(x)]$$

This captures:

- The logarithmic increasing trend
- The oscillatory sinusoidal pattern

This best matches the observed curve.

Option 4:

$$\phi(x) = [1, x, \tan(x)]$$

The tangent function introduces discontinuities and does not match the smooth oscillations in the curve.

Final Answer

$$\boxed{\phi(x) = [1, \log(x), \sin(x)]}$$

Q2 Solution

Given the linear regression model:

$$\hat{y} = 2x$$

The Mean Squared Error (MSE) is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

The dataset is:

$$(x, y) = \{(1, 2), (2, 3), (3, 6)\}$$

Since there are 3 data points, we have:

$$n = 3$$

Compute predictions, for all points

Using $\hat{y} = 2x$:

$$\hat{y}_1 = 2(1) = 2$$

$$\hat{y}_2 = 2(2) = 4$$

$$\hat{y}_3 = 2(3) = 6$$

Compute squared errors

$$(y_1 - \hat{y}_1)^2 = (2 - 2)^2 = 0$$

$$(y_2 - \hat{y}_2)^2 = (3 - 4)^2 = (-1)^2 = 1$$

$$(y_3 - \hat{y}_3)^2 = (6 - 6)^2 = 0$$

Compute MSE

$$\text{MSE} = \frac{1}{3}(0 + 1 + 0)$$

$$\text{MSE} = \frac{1}{3}$$

Compute $3 \times$ MSE

$$3 \times \text{MSE} = 3 \times \frac{1}{3}$$

$$3 \times \text{MSE} = 1$$

Final Answer

$$\boxed{1}$$

Q3. Solution

Given the optimization problem:

$$\min_{w} \, (w - 2)^2 + \lambda w^2$$

where $\lambda > 0$.

Define the objective function

$$f(w) = (w - 2)^2 + \lambda w^2$$

Expand the function, and substitute

$$f(w) = (1 + \lambda)w^2 - 4w + 4$$

Differentiate the function wrt w, and equate it to zero

$$\frac{df(w)}{dw} = 2(1 + \lambda)w - 4$$

Solve for critical point

Set derivative equal to zero:

$$2(1 + \lambda)w - 4 = 0$$

$$w = \frac{2}{1 + \lambda}$$

Verify minimum

Since $\lambda > 0$, the coefficient of w^2 in $f(w)$ is $(1+\lambda) > 0$. Therefore, the function is convex and the critical point is the global minimum.

Final Answer

$$\boxed{w^* = \frac{2}{1+\lambda}}$$

Q4. Solution

The question asks:

Which of the following is the primary purpose of cross-validation in machine learning?

Understand Cross-Validation

Cross-validation is a resampling technique used to evaluate how well a model generalizes to unseen data.

It gives a more reliable estimate of performance compared to a single train-test split.

Let's Evaluate the Options

- **A.** To improve the training accuracy of the model

  Cross-validation does not directly improve training accuracy. It evaluates performance.

- **B.** To obtain an unbiased estimate of the generalization error

  This is correct. Cross-validation estimates how well the model performs on unseen data.

- **C.** To reduce the computational cost of training

  Cross-validation increases computational cost because the model is trained multiple times.

- **D.** To guarantee that the model will not overfit

  Cross-validation helps detect overfitting but does not guarantee prevention.

Final Answer

$$\boxed{\text{To obtain an unbiased estimate of the generalization error}}$$

Q5. Which of the following statements is/are true in Linear Regression settings? **[2 points]**

A. Absolute value loss $|y - \mathbf{w}^\top \mathbf{x}|$ is continuous but not differentiable.

B. Regularisation helps avoid overfitting.

C. A high-degree polynomial fitted to training data also generalizes well to unseen data.

D. K-fold (F-fold) cross-validation benefits by using the whole dataset and is computationally cheaper.

Answer: A,B.

Q6. Which of the following statements is/are true for linear regression (where "linear" means linear in the model parameters)? **[2 points]**

A. $y = w_0 + w_1 x_1 + w_1 w_2 x_2$ is a linear regression model.

B. The solution to $\min_{\mathbf{w}} \|X\mathbf{w} - \mathbf{y}\|_2^2$ is always unique.

C. The squared error is more sensitive to large errors than the absolute error.

D. $y = w_0 + w_1 x_1 + w_2 x_2^2$ is a linear regression model.

Answer: C,D. In option B, for eg. X might have dependent columns and hence multiple solutions might exist. Option A is linear in x but not in model parameters.

Q7. The squared error objective $\|X\mathbf{w} - \mathbf{y}\|_2^2$ in linear regression is **strictly** convex for any matrix X. (True/False)

[1 point]

Answer: False, it is convex but not strictly

Q8. If each data point in a dataset is duplicated before training a linear regression model on the squared error, will the learned model weights change? (Yes/No)

[1 point]

Answer: False, try writing the loss for both cases, it would evaluate to the same expression.